

BIG DATA ASSIGNMENT

1. E-Commerce Big Data

Question:

An e-commerce company receives **5 million transactions per day**. Each transaction generates:

- Transaction record = **4 KB**
- Customer activity/log data = **6 KB**
- Associated metadata = **2 KB**

Calculate:

- a) Total data generated by one transaction.
- b) Total data generated per day.
- c) Data generated in one year.
- d) If the data volume grows by **25% in the next year**, calculate next year's storage requirement.
- e) Identify and explain three Big Data characteristics present in the scenario.

Solution:

a) Total data generated by one transaction

Given:

- Transaction record = 4 KB
- Customer activity/log data = 6 KB
- Metadata = 2 KB

Therefore,

Total data per transaction = $4 + 6 + 2$ KB = 12 KB = $\boxed{12 \text{ KB}}$

Answer: 12 KB per transaction

b) Total data generated per day

Number of transactions per day:

5,000,000 5,000,000

Data per transaction:

12 KB 12\ KB

Therefore,

$5,000,000 \times 12 = 60,000,000 \text{ KB}$ $5,000,000 \times 12 = 60,000,000 \text{ KB}$

Converting KB to GB:

$60,000,000 \times \frac{1}{1024} \approx 57.22 \text{ GB/day}$ $\approx 57.22 \text{ GB/day}$

Answer: Approximately 57.22 GB per day

c) Data generated in one year

Assuming 365 days:

$57.22 \times 365 = 20,885.3 \text{ GB}$ $57.22 \times 365 = 20,885.3 \text{ GB}$

Converting GB to TB:

$20,885.3 \times \frac{1}{1024} \approx 20.4 \text{ TB}$ $\approx 20.4 \text{ TB}$

Answer: Approximately 20.4 TB per year

d) Next year's storage requirement with 25% growth

Current yearly data:

20.4 TB 20.4\ TB

Growth:

25% 25\%

Additional data:

$20.4 \times \frac{25}{100} = 5.1 \text{ TB}$ $20.4 \times \frac{25}{100} = 5.1 \text{ TB}$

Therefore:

$20.4 + 5.1 = 25.5 \text{ TB}$ $20.4 + 5.1 = 25.5 \text{ TB}$

Answer: Approximately 25.5 TB

e) Three Big Data characteristics present in the scenario

1. Volume:

The company generates data from 5 million transactions every day, resulting in approximately 57.22 GB of data daily and more than 20 TB annually.

2. Velocity:

Transaction and customer activity data are generated continuously and at a very high speed. The company must process this data quickly.

3. Variety:

The system handles different types of information, such as transaction records, customer activity logs, and metadata.

2. Distributed Processing Speedup

Question:

A company has **800 GB of log data**. A single computer can process data at **20 GB/hour**.

Calculate:

- a)** Calculate processing time using one computer.
- b)** If the data is equally distributed among 8 computers, calculate the ideal processing time.
- c)** Due to communication and coordination overhead, an additional **30 minutes** is required. Find the actual processing time.
- d)** Calculate the speedup.
- e)** Explain why actual speedup is less than ideal speedup.

Solution:

a) Processing time using one computer

Total data:

800 GB

Processing speed:

20 GB/hour

Therefore,

$$\text{Time} = \frac{800}{20} = 40 \text{ hours} = \boxed{40 \text{ hours}}$$

Answer: 40 hours

b) Ideal processing time using 8 computers

Total data:

$$800 \text{ GB}$$

Number of computers:

$$8$$

Data per computer:

$$\frac{800}{8} = 100 \text{ GB}$$

Each computer processes at 20 GB/hour:

$$\frac{100}{20} = 5 \text{ hours}$$

Answer: 5 hours

c) Actual processing time

Communication and coordination overhead:

$$30 \text{ minutes} = 0.5 \text{ hour}$$

Ideal processing time:

$$5 \text{ hours}$$

Therefore:

$$5 + 0.5 = 5.5 \text{ hours} = \boxed{5.5 \text{ hours}}$$

Answer: 5.5 hours

d) Calculate the speedup

Formula:

$$\text{Speedup} = \frac{\text{Time using one computer}}{\text{Actual distributed time}} = \frac{40}{5.5} = 7.27 \times$$

Answer: Approximately 7.27× speedup

e) Why is actual speedup less than ideal speedup?

The ideal speedup with 8 computers would be:

$8 \times$

However, actual speedup is only **7.27×** because distributed systems have overhead such as:

- Communication between computers
- Data transfer
- Task coordination
- Synchronization
- Scheduling
- Network delays

Therefore, some processing time is spent on activities other than actual computation.

3. Distributed Storage with Replication

Question:

An organization has **60 TB of data** that must be stored in a distributed file system. Each storage node provides **8 TB usable capacity**. For reliability, every piece of data is maintained in **3 copies**.

Calculate:

- a) Storage required without replication.
- b) Storage required with three copies.
- c) Minimum number of storage nodes required.
- d) Calculate the additional storage overhead caused by replication.
- e) Explain why replication is important in a distributed file system.

Solution:

a) Storage required without replication

Original data:

60 TB\boxed{60\ TB}

Answer: 60 TB

b) Storage required with three copies

Replication factor:

3

Therefore:

$60 \times 3 = 180$ TB $60 \times 3 = 180\ TB$

Answer: 180 TB

c) Minimum number of storage nodes required

Each node provides:

8 TB $8\ TB$

Total required storage:

180 TB $180\ TB$

Number of nodes:

$180 \div 8 = 22.5$ $\frac{180}{8} = 22.5$

Since a fraction of a node is not possible, we round up:

23 nodes\boxed{23\ nodes}

Answer: Minimum 23 storage nodes

d) Additional storage overhead caused by replication

Storage with replication:

180 TB $180\ TB$

Original storage:

60 TB60\ TB

Additional storage:

$180 - 60 = 120$ TB $180 - 60 = \boxed{120}$ TB}

Therefore, replication creates **120 TB of additional storage requirement**.

e) Importance of replication

Replication means keeping multiple copies of the same data on different storage nodes.

It is important because:

1. **Fault tolerance:** If one node fails, another copy can be used.
 2. **Data availability:** Data remains accessible even when a node is unavailable.
 3. **Data protection:** It reduces the risk of permanent data loss.
 4. **Improved performance:** Multiple copies can sometimes allow faster data access.
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4. Hospital Big Data

Question:

A hospital generates:

- **10,000 patient records/day × 50 KB**
- **2,000 medical images/day × 20 MB**
- **500 GB CCTV data/day**
- **100 GB sensor data/day**

Calculate:

- a) Calculate patient-record data generated per day.
- b) Calculate medical-image data per day.
- c) Calculate total approximate daily data generation.
- d) Calculate storage required for 30 days.
- e) Classify the data as structured/unstructured and identify relevant Big Data characteristics.

Solution:

a) Patient-record data generated per day

Number of patient records:

10,000 10,000

Size of each record:

50 KB 50\ KB

Therefore:

$10,000 \times 50 = 500,000$ KB $10,000 \times 50 = 500,000$ \ KB

Convert to MB:

$500,000 / 1024 = 488.28$ MB $\frac{500,000}{1024} = 488.28$ \ MB

Convert to GB:

$488.28 / 1024 \approx 0.477$ GB $\frac{488.28}{1024} \approx 0.477$ \ GB

Answer: Approximately 488.28 MB or 0.477 GB per day

b) Medical-image data per day

Number of images:

2,000 2,000

Size of each image:

20 MB 20\ MB

Therefore:

$2,000 \times 20 = 40,000$ MB $2,000 \times 20 = 40,000$ \ MB

Convert to GB:

$40,000 / 1024 = 39.06$ GB $\frac{40,000}{1024} = 39.06$ \ GB

Answer: Approximately 39.06 GB per day

c) Total approximate daily data generation

Daily data:

- Patient records = 0.477 GB
- Medical images = 39.06 GB
- CCTV = 500 GB

- Sensor data = 100 GB

Therefore:

$$0.477 + 39.06 + 500 + 1000.477 + 39.06 + 500 + 100 = 639.537 \text{ GB} = \boxed{639.537 \text{ GB}}$$

Approximately:

$$639.54 \text{ GB/day} = \boxed{639.54 \text{ GB/day}}$$

or approximately:

$$0.625 \text{ TB/day} = \boxed{0.625 \text{ TB/day}}$$

d) Storage required for 30 days

$$639.54 \times 30 = 19,186.2 \text{ GB} = 19,186.2 \text{ GB}$$

Convert to TB:

$$19,186.2 \div 1024 \approx 18.74 \text{ TB} \approx \boxed{18.74 \text{ TB}}$$

Answer: Approximately 18.74 TB for 30 days

e) Classification and Big Data characteristics

Data	Type
Patient records	Structured
Medical images	Unstructured
CCTV data	Unstructured
Sensor data	Structured/Semi-structured

Big Data characteristics:

1. Volume:

The hospital generates approximately 640 GB of data every day.

2. Velocity:

CCTV and sensor data are generated continuously and rapidly.

3. Variety:

The hospital handles different forms of data, including patient records, images, videos, and sensor information.

5. Vertical vs Horizontal Scaling

Question:

A server currently processes **1 TB of data in 8 hours**.

Option A: Upgrade the server so that its processing capability doubles at a cost of **₹8 lakh**.

Option B: Use four identical existing processing nodes costing **₹2 lakh each**. Assume perfect workload distribution initially.

Calculate:

- a) Find the processing time after vertical scaling.
- b) Find the ideal processing time with four nodes.
- c) If distributed-system overhead is 30 minutes, calculate actual processing time.
- d) Calculate actual speedup.
- e) Based on scalability and processing time, discuss vertical vs horizontal scaling.

Solution:

a) Processing time after vertical scaling

Original processing time:

8 hours

Processing capability doubles.

Therefore, processing time becomes half:

$$8 \div 2 = 4 \text{ hours}$$

Answer: 4 hours

b) Ideal processing time with four nodes

Original processing time:

8 hours

Number of nodes:

4

Assuming perfect workload distribution:

$$84 = 2 \text{ hours} \frac{8}{4} = \boxed{2 \text{ hours}}$$

Answer: 2 hours

c) Actual processing time

Overhead:

$$30 \text{ minutes} = 0.5 \text{ hour} \quad 30 \text{ minutes} = 0.5 \text{ hour}$$

Ideal processing time:

$$2 \text{ hours} \quad 2 \text{ hours}$$

Therefore:

$$2 + 0.5 = 2.5 \text{ hours} \quad 2 + 0.5 = \boxed{2.5 \text{ hours}}$$

Answer: 2.5 hours

d) Actual speedup

Formula:

$$\text{Speedup} = \frac{\text{Original time}}{\text{Actual time}} \quad \text{Speedup} = \frac{8}{2.5} = 3.2 \times = \boxed{3.2 \times}$$

Answer: 3.2×

e) Vertical vs Horizontal Scaling

Feature	Vertical Scaling	Horizontal Scaling
Method	Upgrade existing server	Add more servers
Processing time	4 hours	2.5 hours actual
Scalability	Limited	High
Fault tolerance	Lower	Higher
Expansion	Hardware upgrade	Add more nodes

Cost	₹8 lakh	₹8 lakh
Suitable for Big Data	Less suitable	More suitable

Conclusion:

Vertical scaling improves the power of one machine, but it has hardware limitations. Horizontal scaling adds multiple machines and allows workload distribution. For Big Data applications, **horizontal scaling is generally more scalable and fault tolerant.**

6. Data Growth Forecasting

Question:

A company currently stores **100 TB of data**. Its data volume increases by **30% every year**.

Calculate:

- Calculate storage required after Year 1.
- Calculate storage required after Year 2.
- Calculate storage required after Year 3.
- If each node has **20 TB usable capacity**, calculate nodes required after three years.
- Explain how distributed systems support such growth.

Solution:

Growth rate:

30%

Growth multiplier:

$1 + 0.30 = 1.30$

a) Storage required after Year 1

$100 \times 1.30 = 130 \text{ TB}$

Answer: 130 TB

b) Storage required after Year 2

$130 \times 1.30 = 169 \text{ TB}$

Answer: 169 TB

c) Storage required after Year 3

$$169 \times 1.30169 \times 1.30 = 219.7 \text{ TB} = \boxed{219.7 \text{ TB}}$$

Answer: 219.7 TB

d) Number of nodes required after three years

Data after three years:

$$219.7 \text{ TB}$$

Capacity of each node:

$$20 \text{ TB}$$

Number of nodes:

$$\frac{219.7}{20} = 10.985$$

Since we need complete nodes:

$$11 \text{ nodes} = \boxed{11 \text{ nodes}}$$

Answer: 11 nodes

e) How distributed systems support data growth

Distributed systems support increasing data volumes by:

1. **Adding more storage nodes** as data increases.
 2. **Distributing data** across multiple machines.
 3. **Providing fault tolerance** using replication.
 4. **Supporting parallel processing** across multiple nodes.
 5. **Providing scalability** without replacing one large server.
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7. Node Failure & Data Availability

Question:

A distributed file system contains **40 storage nodes**, each storing **5 TB**. The system is currently using **70% of its raw storage capacity**.

Calculate:

- a) Calculate total raw storage capacity.
- b) Calculate currently used storage.
- c) Calculate available free storage.
- d) If 4 nodes fail, calculate the raw storage capacity remaining online.
- e) Discuss how node failures can affect availability and how replication can address the problem.

Solution:

a) Total raw storage capacity

Number of nodes:

40

Capacity per node:

5 TB

Therefore:

$$40 \times 5 = 200 \text{ TB}$$

Answer: 200 TB

b) Currently used storage

Used percentage:

70%

Therefore:

$$200 \times \frac{70}{100} = 140 \text{ TB}$$

Answer: 140 TB

c) Available free storage

$$200 - 140 = 60 \text{ TB}$$

Answer: 60 TB

d) Raw storage capacity after 4 nodes fail

Capacity of 4 nodes:

$$4 \times 5 = 20 \text{ TB}$$

Remaining capacity:

$$200 - 20 = 180 \text{ TB}$$

Answer: 180 TB

e) Node failures and replication

In a distributed file system, failure of a storage node can make data unavailable if that node contains the only copy of the data.

Replication solves this problem by maintaining multiple copies of data on different nodes.

For example, if a file has three copies and one node fails, the remaining two copies can still be accessed.

Therefore, replication provides:

- **Fault tolerance**
 - **High availability**
 - **Data protection**
 - **Reduced risk of data loss**
 - **Continuous operation during node failures**
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8. Social-Media Big Data

Question:

A social-media company receives data from **2 million active users**. Each user generates an average of **5 MB of images/videos** and **500 KB of textual/log data per day**. The company operates a distributed cluster in which each storage node provides **5 TB usable capacity**.

Calculate and analyze:

- a) Total data generated by one user per day.
- b) Total data generated by all users per day.

- c) Storage required for 30 days.
- d) Determine the approximate number of 5 TB storage nodes required, ignoring replication.
- e) Identify the types of Big Data generated and explain any three Big Data characteristics evident in this application.

Solution:

a) Total data generated by one user per day

Images/videos:

5 MB

Textual/log data:

500 KB

Convert KB to MB:

$$500 \times \frac{1}{1024} = 0.488 \text{ MB}$$

Total:

$$5 + 0.488 = 5.488 \text{ MB}$$

Approximately:

$$5.49 \text{ MB/user/day}$$

b) Total data generated by all users per day

Number of users:

2,000,000

Data per user:

5.488 MB

Therefore:

$$2,000,000 \times 5.488 = 10,976,000 \text{ MB}$$

Convert to GB:

$$\frac{10,976,000}{1024} = 10,718.75 \text{ GB}$$

Convert to TB:

$$\frac{10,718.75}{1024} \approx 10.47 \text{ TB/day}$$

Answer: Approximately 10.47 TB per day

c) Storage required for 30 days

Daily data:

10.47 TB

For 30 days:

$10.47 \times 30 = 314.1 \text{ TB}$

Using the more precise value:

$\approx 314.06 \text{ TB}$

Answer: Approximately 314.06 TB

d) Number of 5 TB storage nodes required

Total storage:

314.06 TB

Capacity per node:

5 TB

Number of nodes:

$\frac{314.06}{5} = 62.812$

Since we need complete nodes:

63 nodes

Answer: Approximately 63 storage nodes

This calculation ignores replication, as specified in the question.

e) Types of Big Data and three Big Data characteristics

Types of data generated:

1. Images and videos:

These are generally **unstructured data** because they do not follow a fixed tabular format.

2. Textual data:

Posts, comments, messages, and other text can be structured or semi-structured depending on how they are stored.

3. Log data:

User activity logs are generally **semi-structured data**, often containing fields such as user ID, timestamp, action, and device information.

Three Big Data characteristics:

1. Volume:

2 million users generate approximately 10.47 TB of data every day, resulting in more than 300 TB in just 30 days.

2. Velocity:

Social-media data is generated continuously as users upload images, videos, comments, and messages.

3. Variety:

The platform handles multiple data types, including images, videos, text, and user activity logs.

Overall Conclusion

The above problems demonstrate the major concepts of **Big Data and Distributed Systems**, including:

- Large-scale data generation
- Data growth forecasting
- Distributed processing
- Processing speedup
- Distributed storage
- Replication
- Node failures
- Fault tolerance
- Vertical and horizontal scaling
- Structured, semi-structured, and unstructured data
- Big Data characteristics such as **Volume, Velocity, and Variety**